

Ultimate checklist for SDM building – student compilation

This is a checklist compiled by students at the Science School on Quantitative Ecology on the 14th of September. It contains aspects that should be considered to create scientifically sound and ecologically meaningful species distribution models.

Data

- Spatial and temporal scale: choose the one best fitting your research question or hypotheses
- Use high-quality data and know the source of your data
- Check for spatial bias and autocorrelation
- Check taxonomy: your study species might be known under different synonyms
- Check for data gaps and NA values

Species data

- Filter and clean occurrence data
- If using citizen science data: check the data properly. Are there biases?

Environmental data

- Select ecologically meaningful variables
- Make sure your variables all have the same resolution, projection and extent. The overlay needs to be perfect.
- Check correlation between parameters and remove one of highly correlated variables.

Model building

- Choose algorithm fit for purpose (species characteristic and hypotheses)
- Fine-tune model (don't only use default setting)

Model quality check

- Split data in training and test data (for example by building spatial blocks)
- Does output make ecologically sense? (visualization is key)

Projection

- Create map with proper legend, orientation, coordinates

Visualization

- Choose if gradient or binary map is more suitable
- Select colours from colour-blind friendly pallet
- Choose fit-for-purpose target area

Reporting

- Share your data (FAIRE and SHARE principle)
- Produce commented and reproducible code